

Engineering Curves: Construction of Epicycloid

Rolling Circle Method with Tangent & Normal

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Problem Statement

Problem

A generating circle of diameter 30 mm rolls externally without slipping on a directing circle of diameter 90 mm. Construct one complete epicycloid arch corresponding to $\angle AOB = 120^\circ$, divide the arch into 12 equal angular parts, and retain every construction element in all subsequent views.

Given Data and Calculations

Given data

Diameter of directing circle = 90 mm,

$$R = 45 \text{ mm},$$

Diameter of generating circle = 30 mm,

$$r = 15 \text{ mm},$$

$$\angle AOB = 120^\circ,$$

Let number of divisions = 12.

Calculations

$$OC_0 = R + r = 45 + 15 = 60 \text{ mm},$$

$$AC_0 = r = 15 \text{ mm},$$

$$\Delta\theta = \frac{120^\circ}{12} = 10^\circ.$$

Generating-circle division

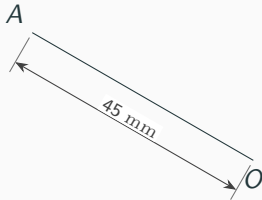
$$\text{Division angle} = \frac{360^\circ}{12} = 30^\circ.$$

Rolling relation

Each 10° centre step $\leftrightarrow$ 30° generating-circle step.

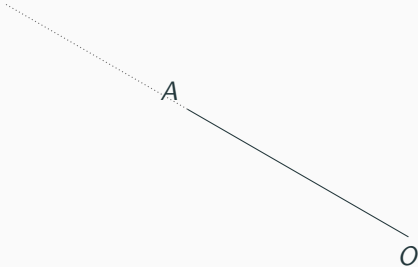
Step 1: Draw OA

$OA = 45 \text{ mm.}$



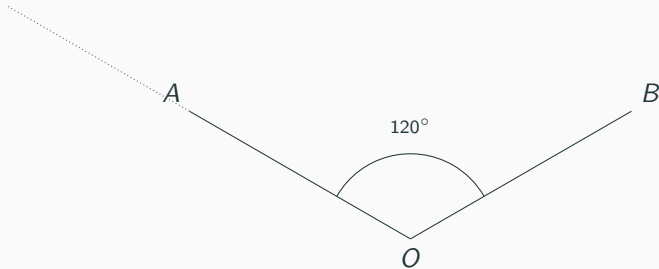
Step 2: Extend OA

Extend OA .



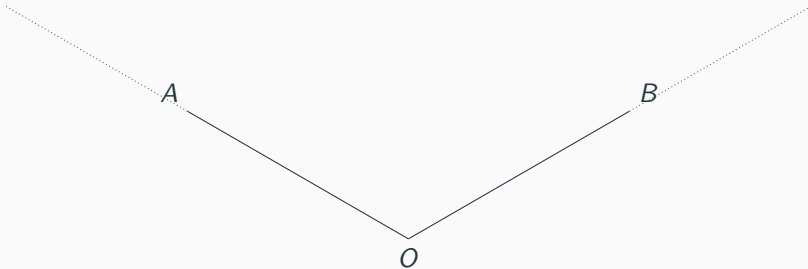
Step 3: Draw OB at 120°

$$\angle AOB = 120^\circ.$$



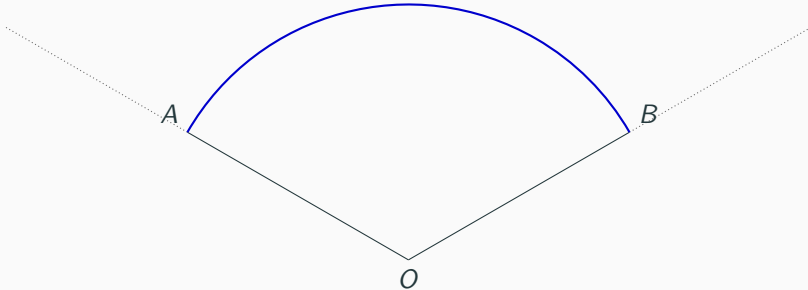
Step 4: Extend OB

Extend OB .



Step 5: Draw the directing arc AB

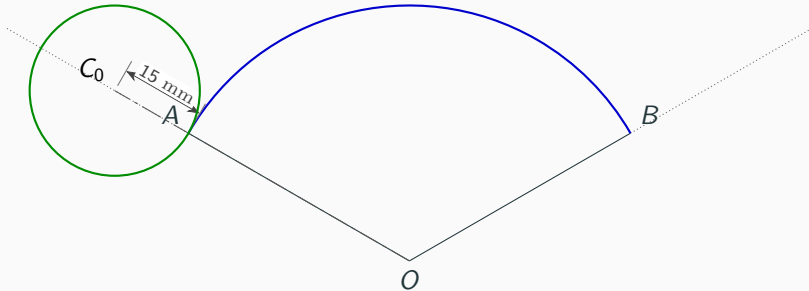
Blue arc AB : directing circle.



Step 6: Locate C_0 and the generating circle

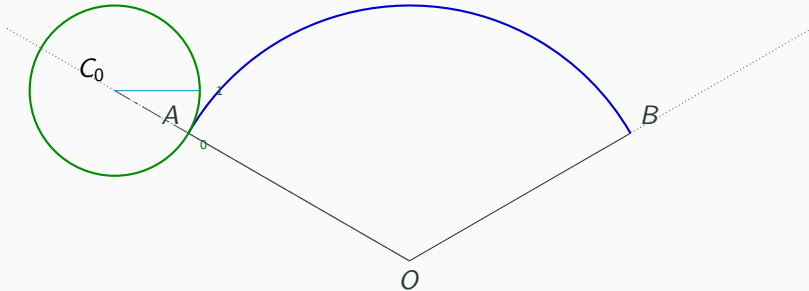
$AC_0 = 15 \text{ mm.}$

Green: generating circle.



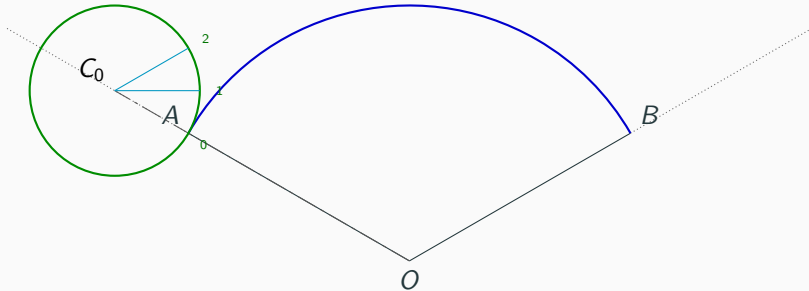
Step 7: Divide the generating circle into 12 equal parts

Green: generating circle.
30° divisions.



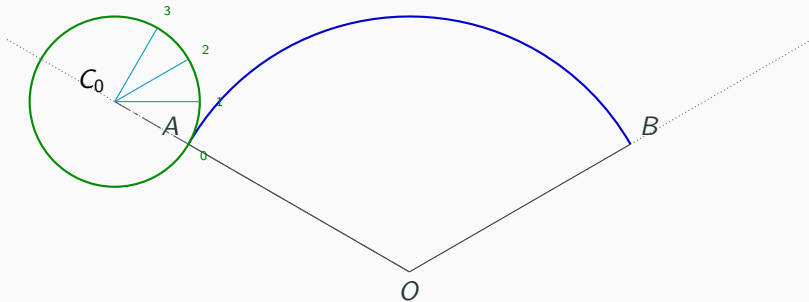
Step 7: Divide the generating circle into 12 equal parts

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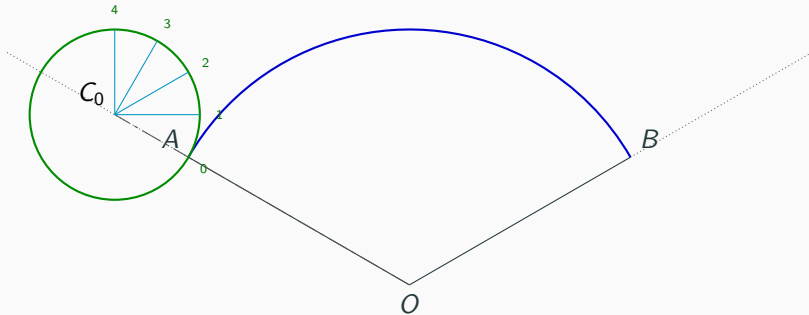
Step 7: Divide the generating circle into 12 equal parts

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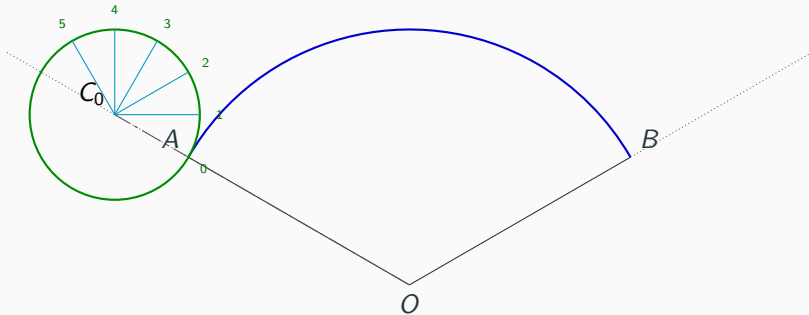
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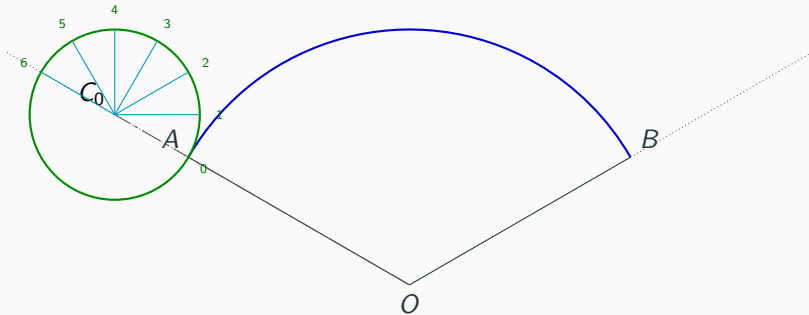
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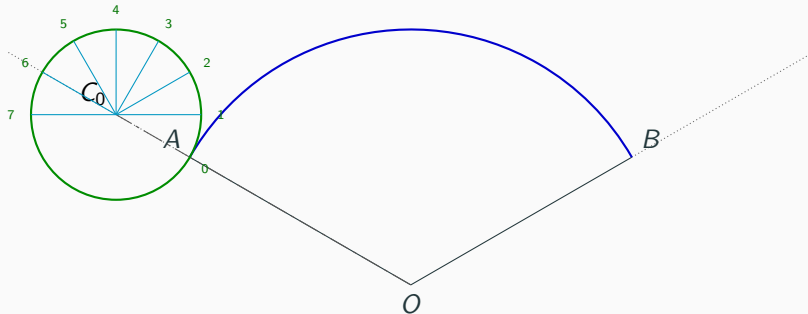
Step 7: Divide the generating circle into 12 equal parts

Green: generating circle.
30° divisions.



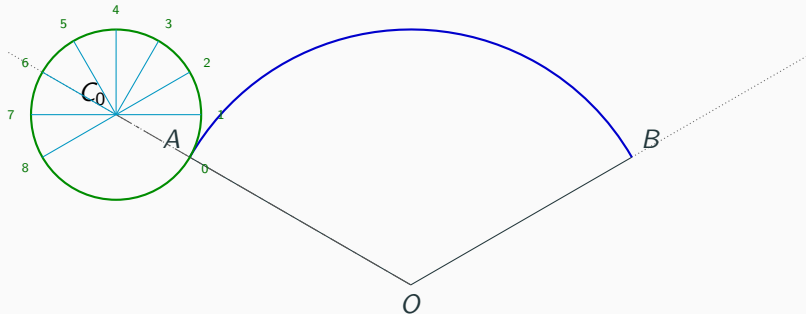
Step 7: Divide the generating circle into 12 equal parts

Green: generating circle.
30° divisions.



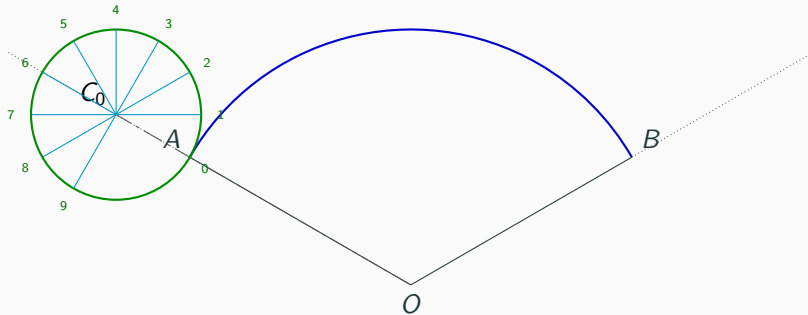
Step 7: Divide the generating circle into 12 equal parts

Green: generating circle.
30° divisions.



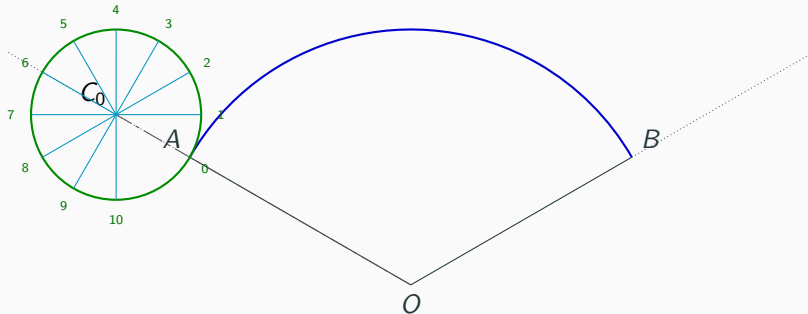
Step 7: Divide the generating circle into 12 equal parts

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30° divisions.



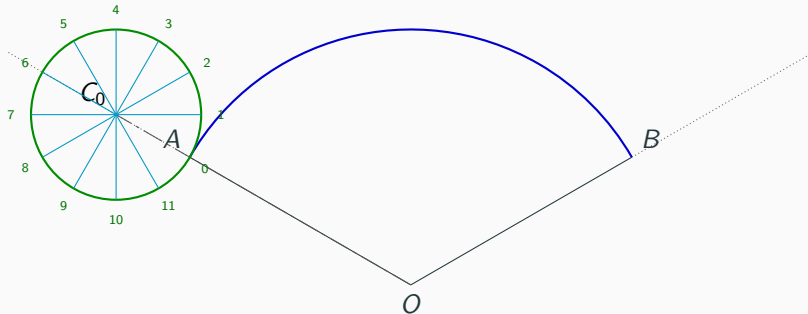
Step 7: Divide the generating circle into 12 equal parts

Green: generating circle.
30° divisions.



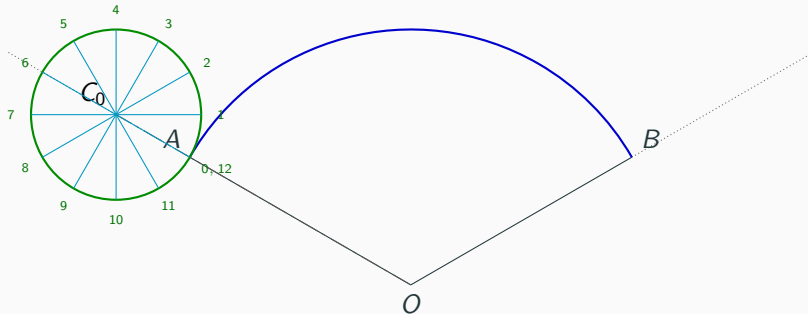
Step 7: Divide the generating circle into 12 equal parts

Green: generating circle.
30° divisions.



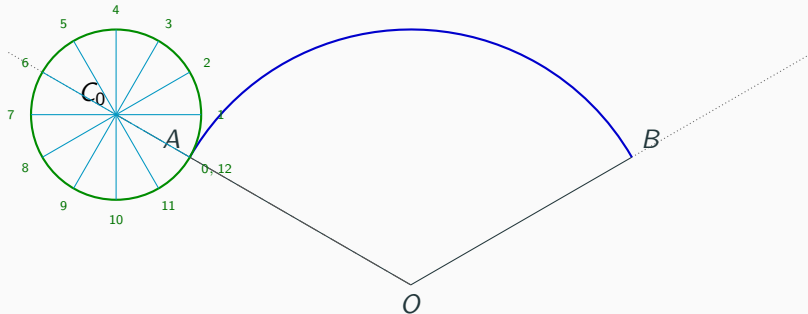
Step 7: Divide the generating circle into 12 equal parts

Green: generating circle.
30° divisions.



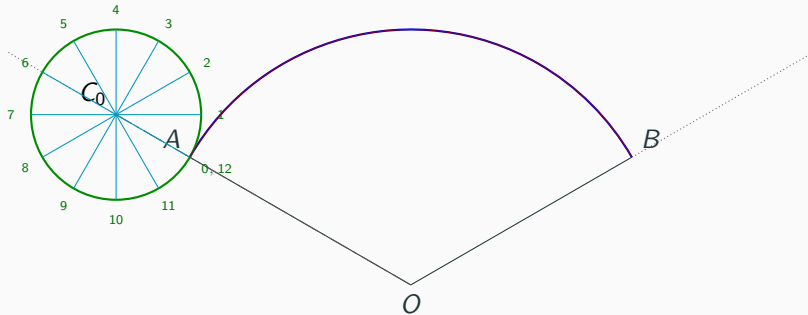
Step 8: Complete the generating-circle divisions

Generating-circle divisions.



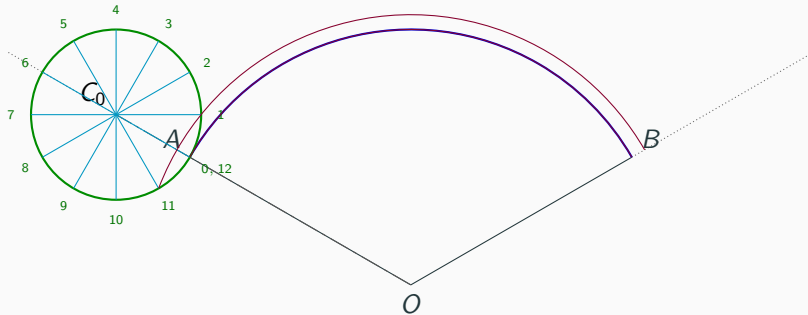
Step 9: Draw concentric construction arcs

Concentric construction arc.



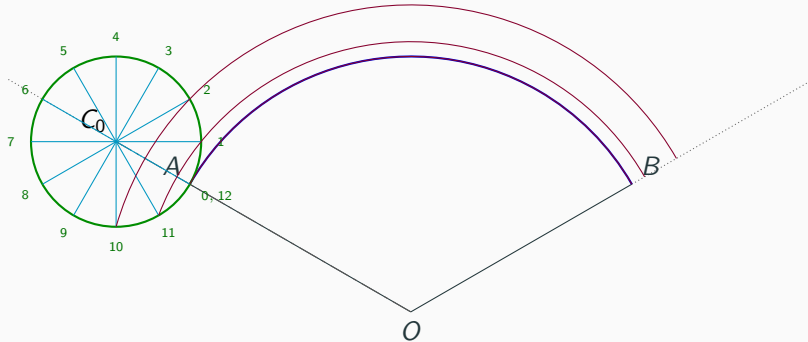
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Concentric construction arc.



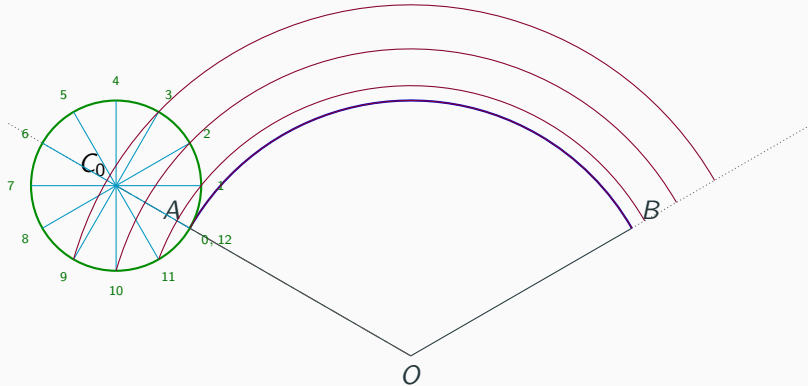
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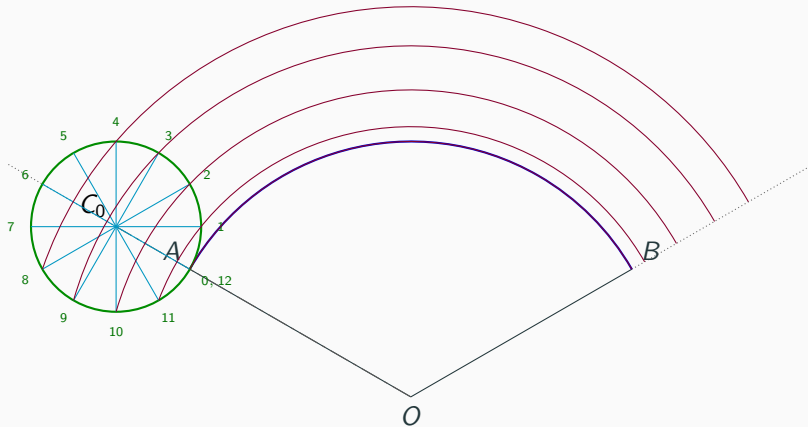
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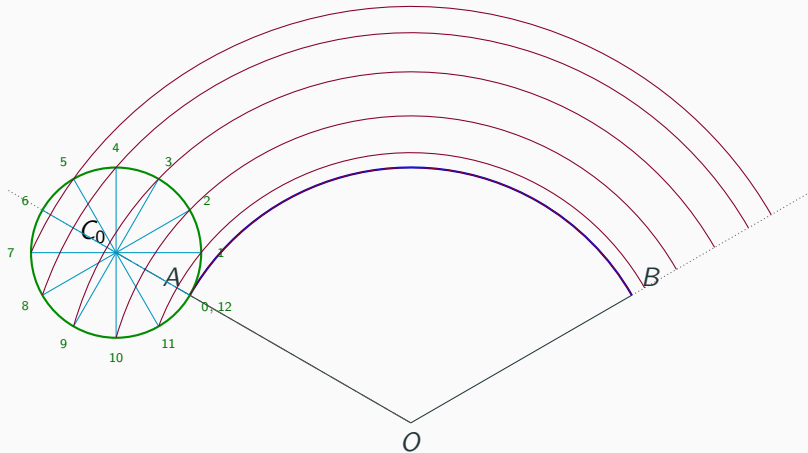
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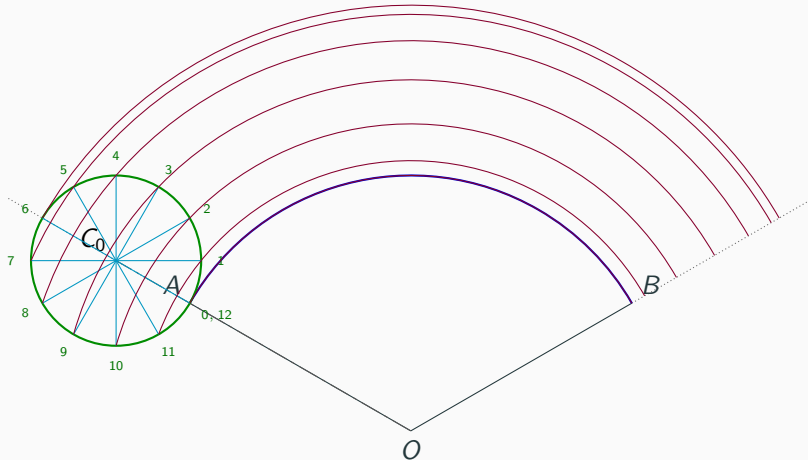
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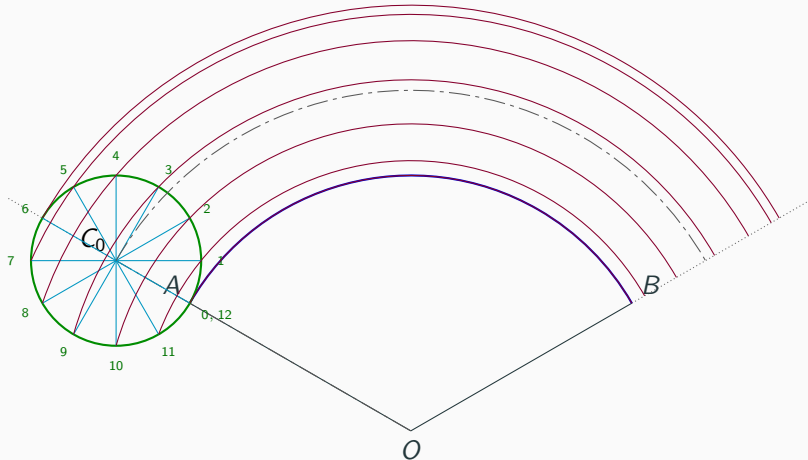
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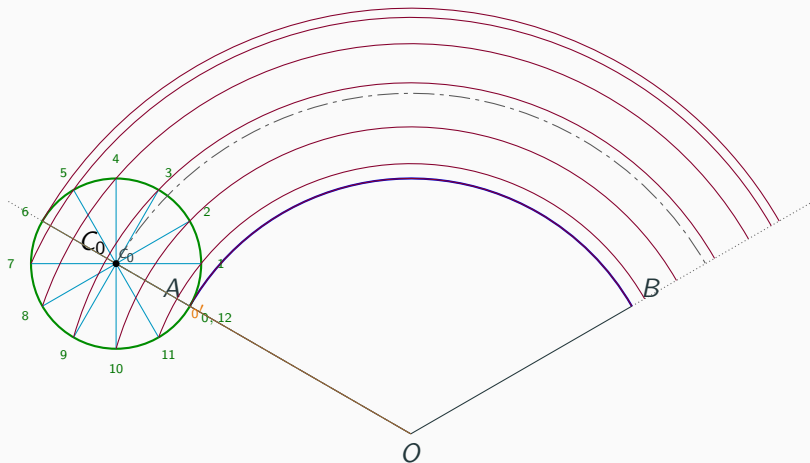
Step 10: Draw the centre arc through C_0

Centre curve: ISO axis line.



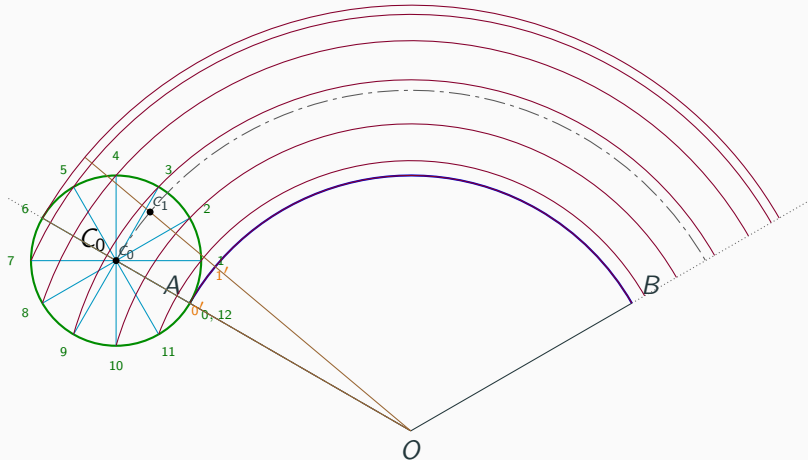
Step 11: Divide $\angle AOB$ into 12 equal parts

10° divisions.



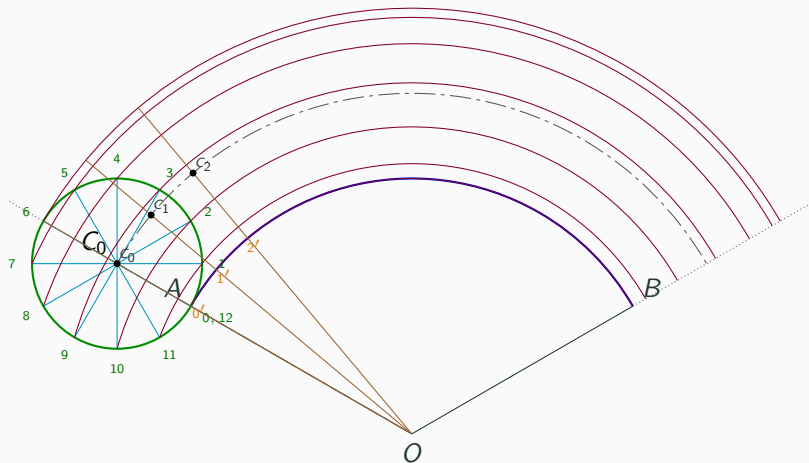
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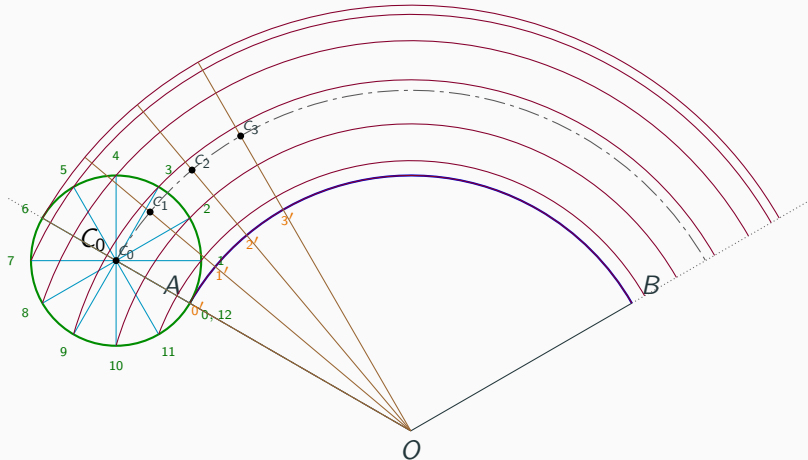
Step 11: Divide $\angle AOB$ into 12 equal parts

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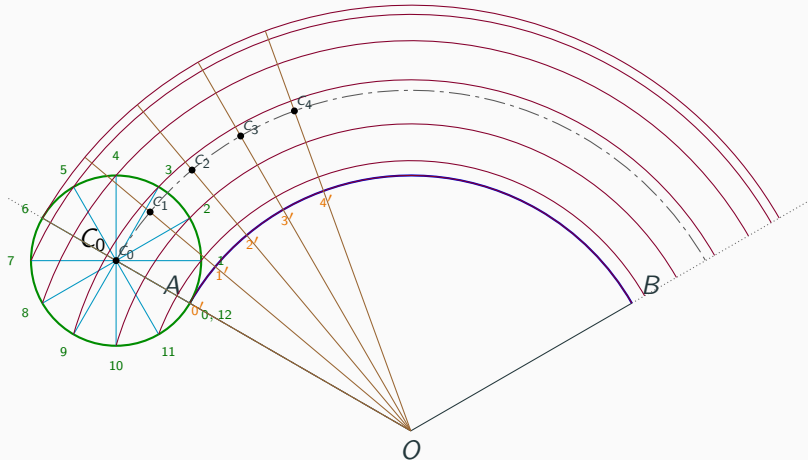
Step 11: Divide $\angle AOB$ into 12 equal parts

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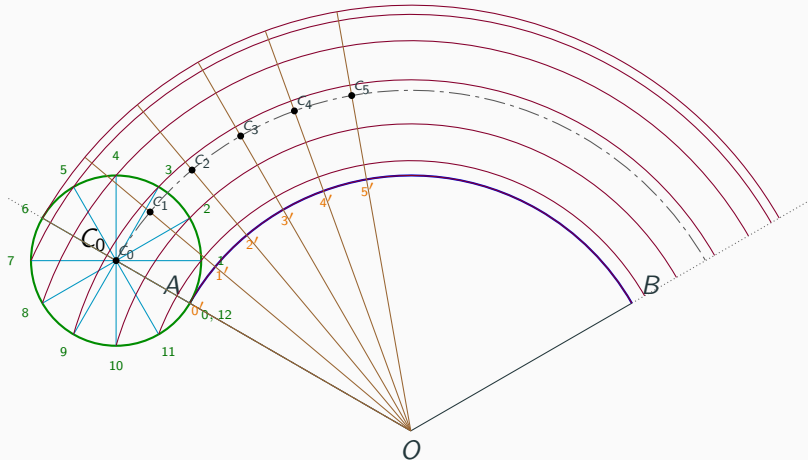
Step 11: Divide $\angle AOB$ into 12 equal parts

10° divisions.



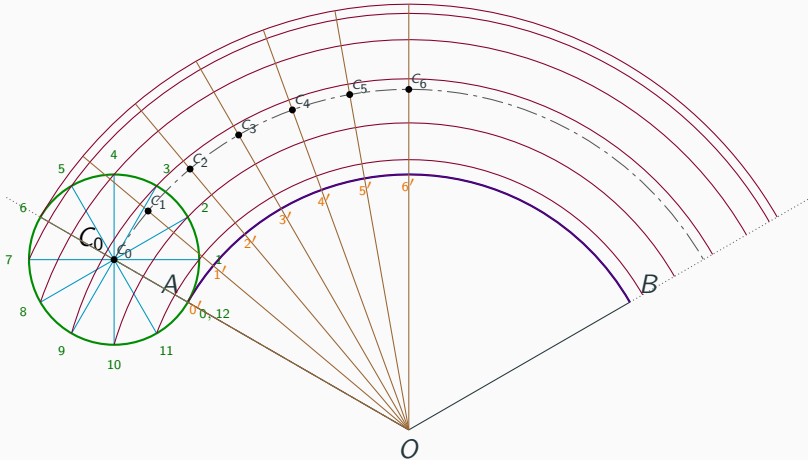
Step 11: Divide $\angle AOB$ into 12 equal parts

10° divisions.



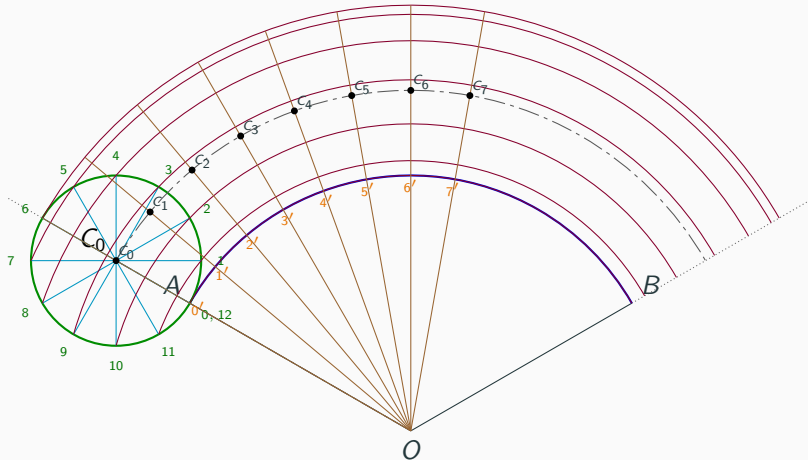
Step 11: Divide $\angle AOB$ into 12 equal parts

10^0 divisions.



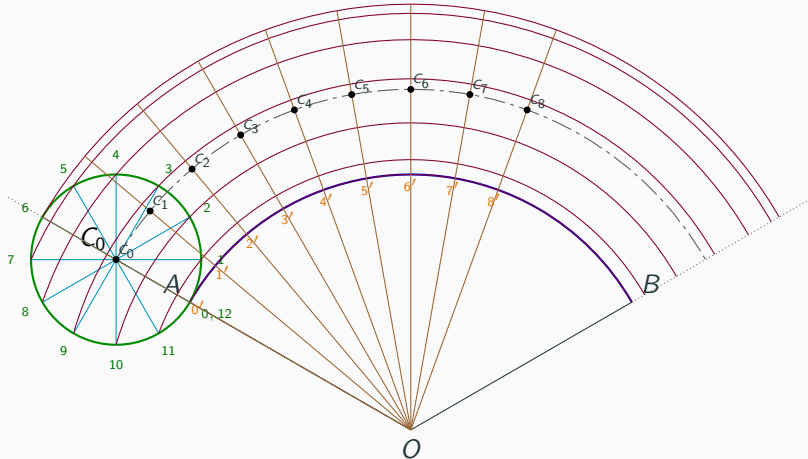
Step 11: Divide $\angle AOB$ into 12 equal parts

10° divisions.



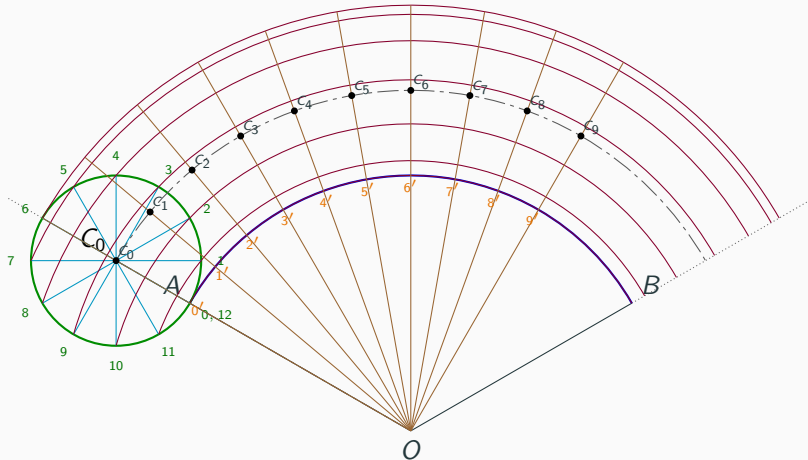
Step 11: Divide $\angle AOB$ into 12 equal parts

10° divisions.



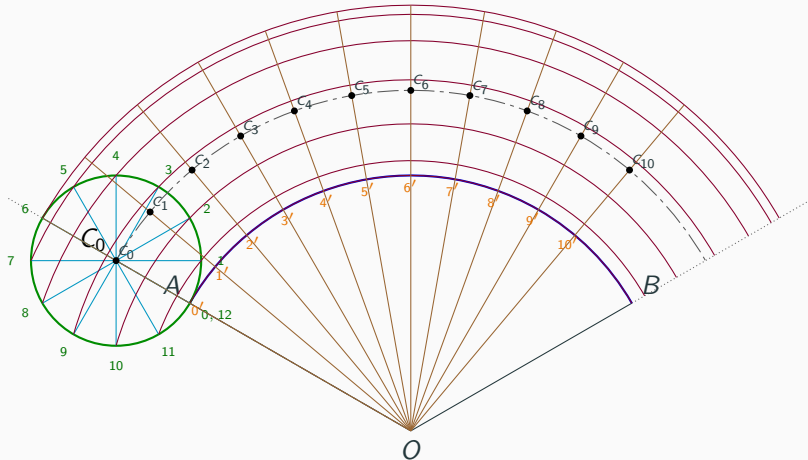
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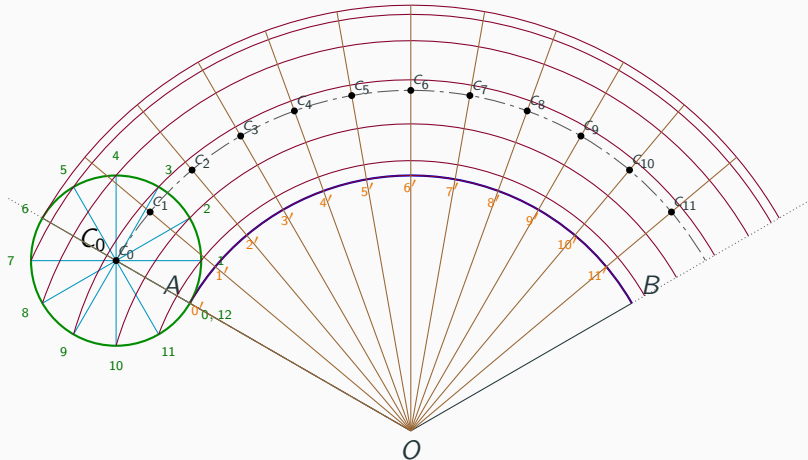
Step 11: Divide $\angle AOB$ into 12 equal parts

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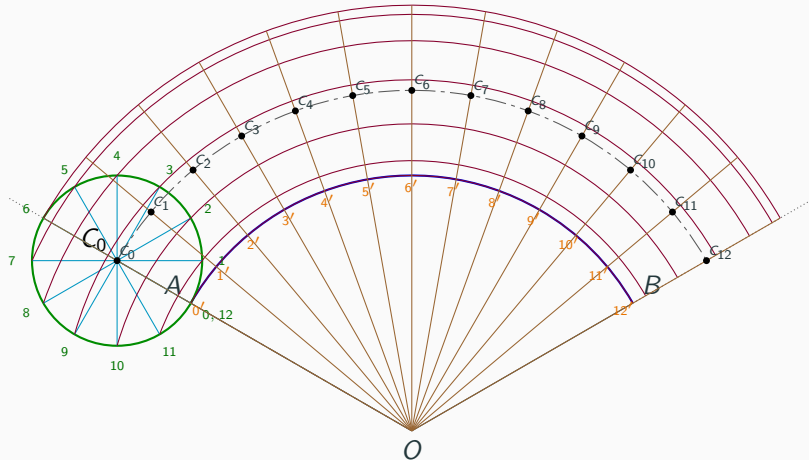
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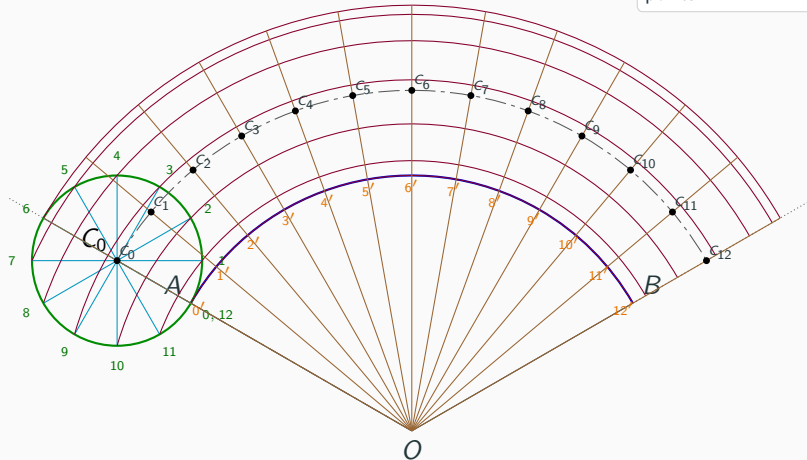
Step 11: Divide $\angle AOB$ into 12 equal parts

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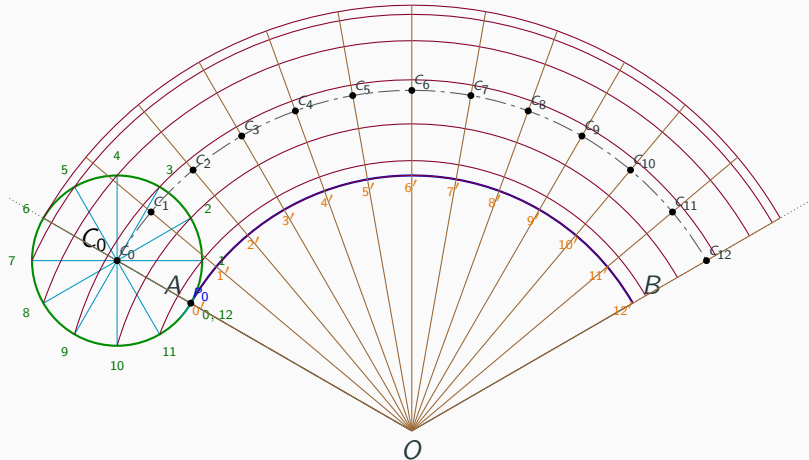
Step 12: Complete the angular divisions and points

Division points and centre points.



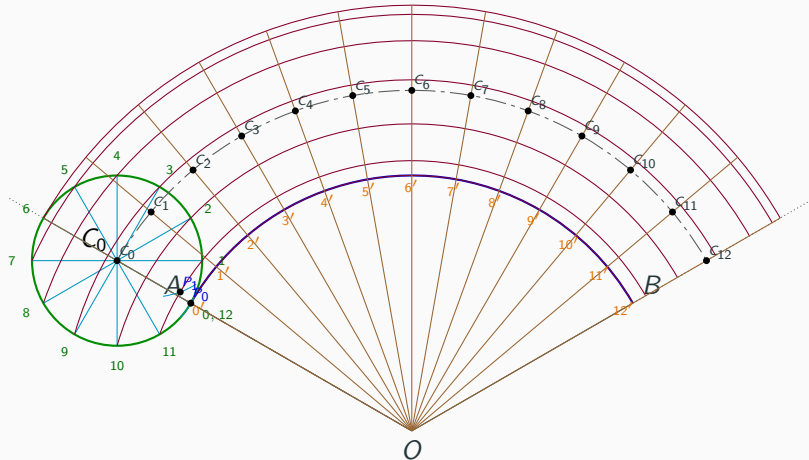
Step 13: Draw the small cutting arc and locate P_i

Cyan arc: radius 15 mm.



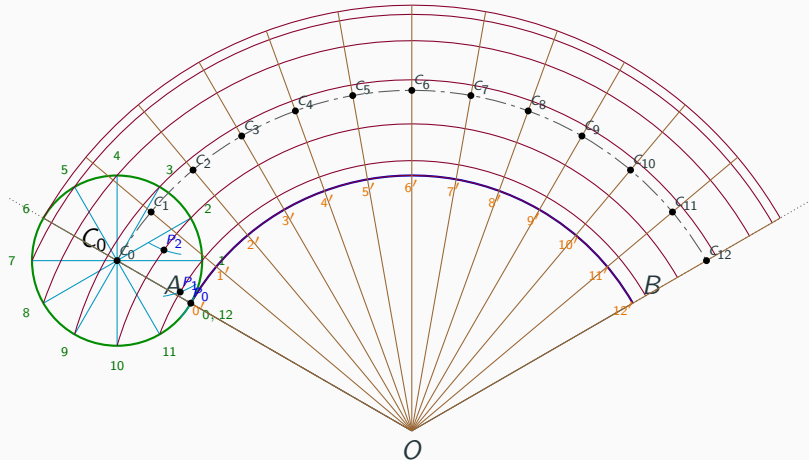
Step 13: Draw the small cutting arc and locate P_i

Cyan arc: radius 15 mm.



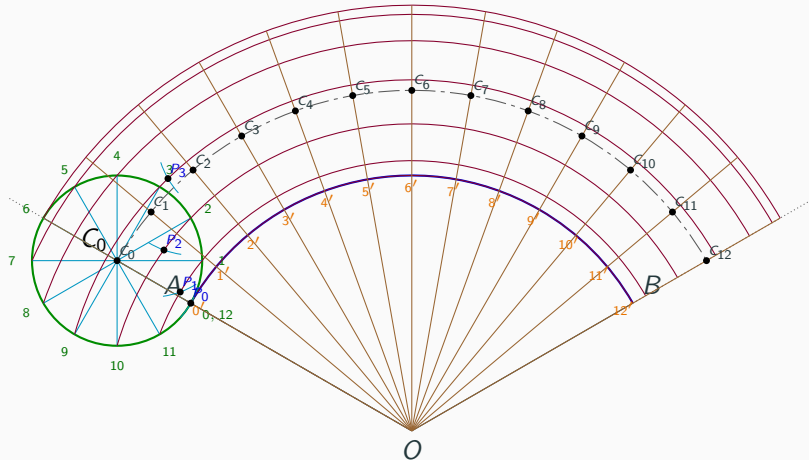
Step 13: Draw the small cutting arc and locate P_i

Cyan arc: radius 15 mm.



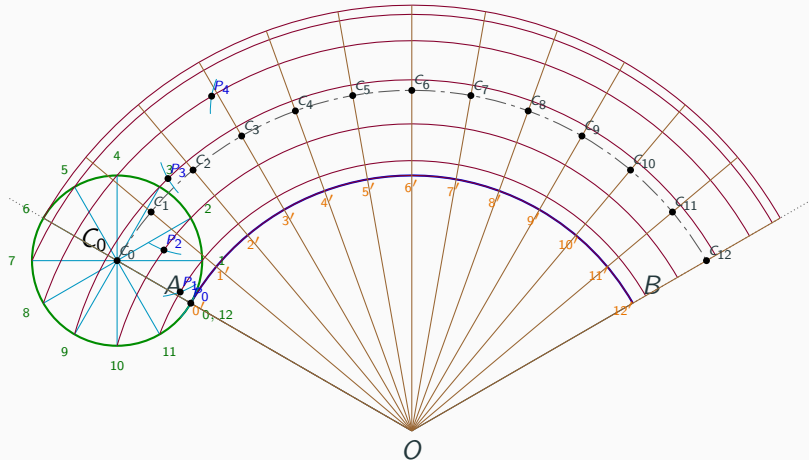
Step 13: Draw the small cutting arc and locate P_i

Cyan arc: radius 15 mm.



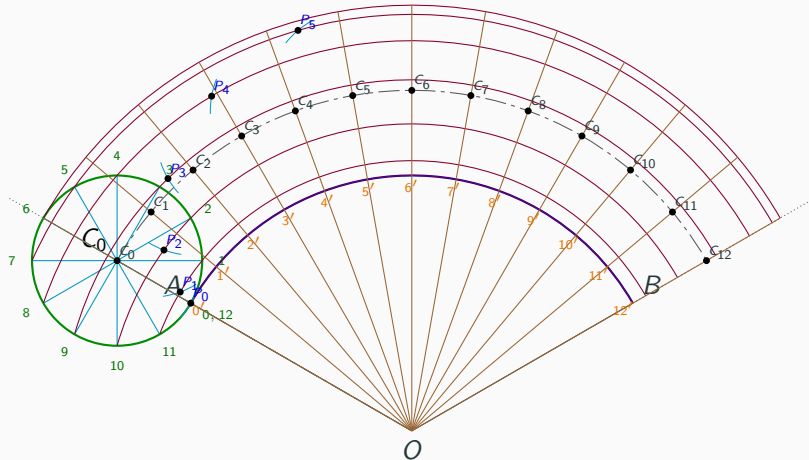
Step 13: Draw the small cutting arc and locate P_i

Cyan arc: radius 15 mm.

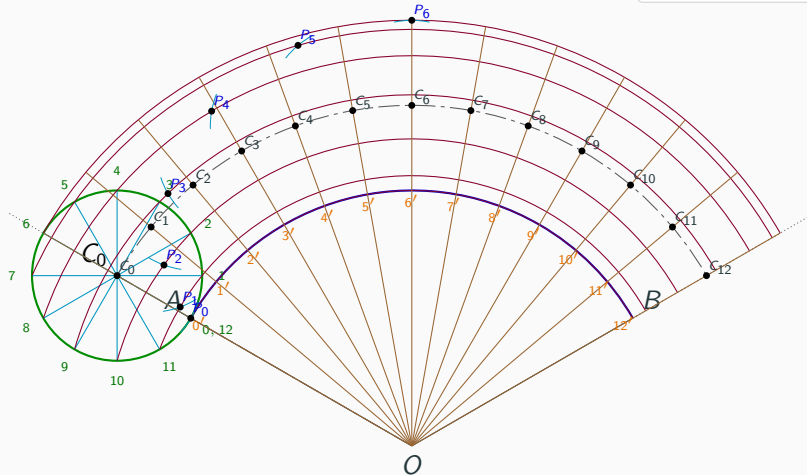


Step 13: Draw the small cutting arc and locate P_i

Cyan arc: radius 15 mm.

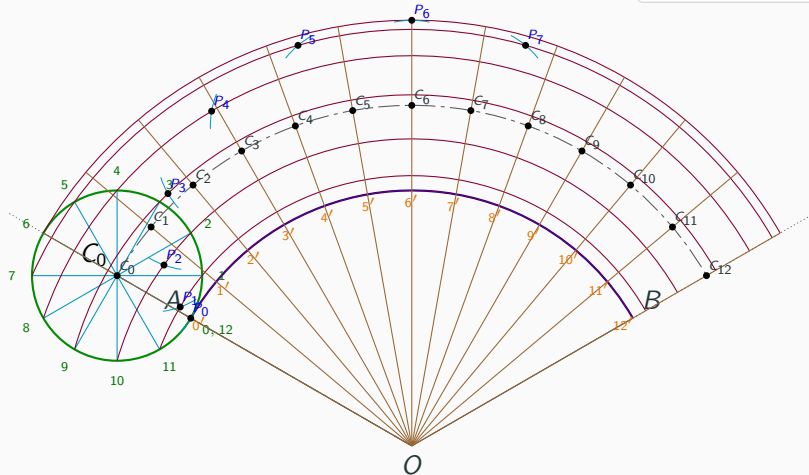


Step 13: Draw the small cutting arc and locate P_i



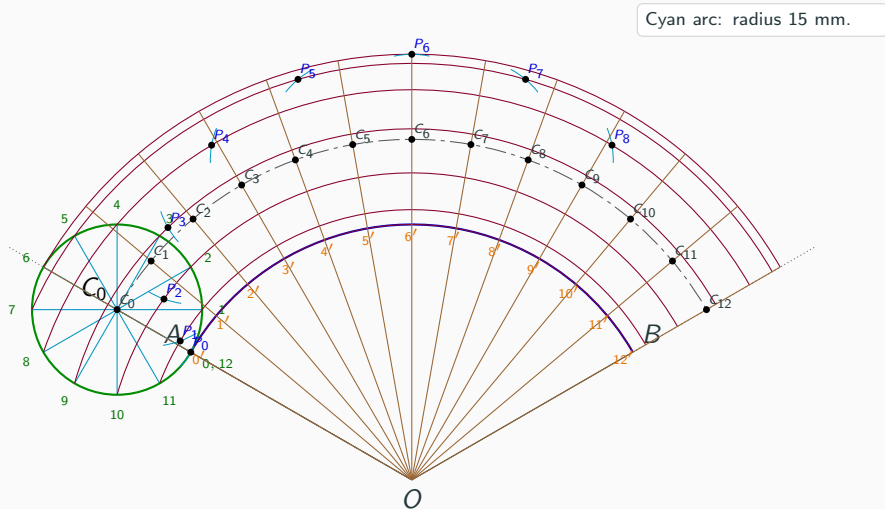
Cyan arc: radius 15 mm.

Step 13: Draw the small cutting arc and locate P_i

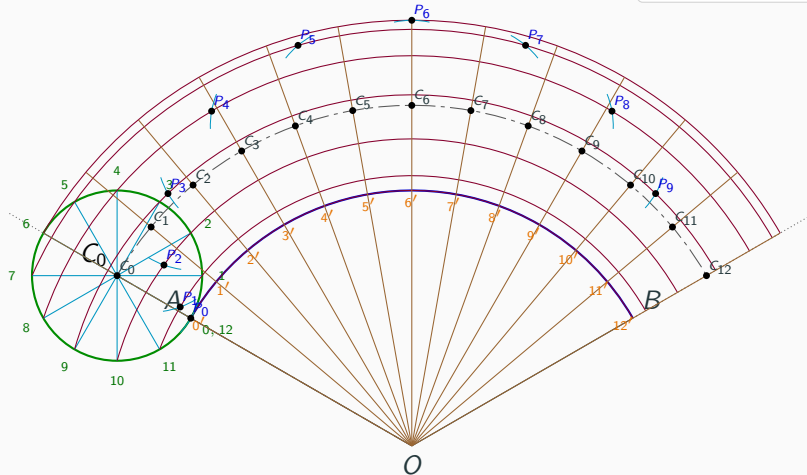


Cyan arc: radius 15 mm.

Step 13: Draw the small cutting arc and locate P_i

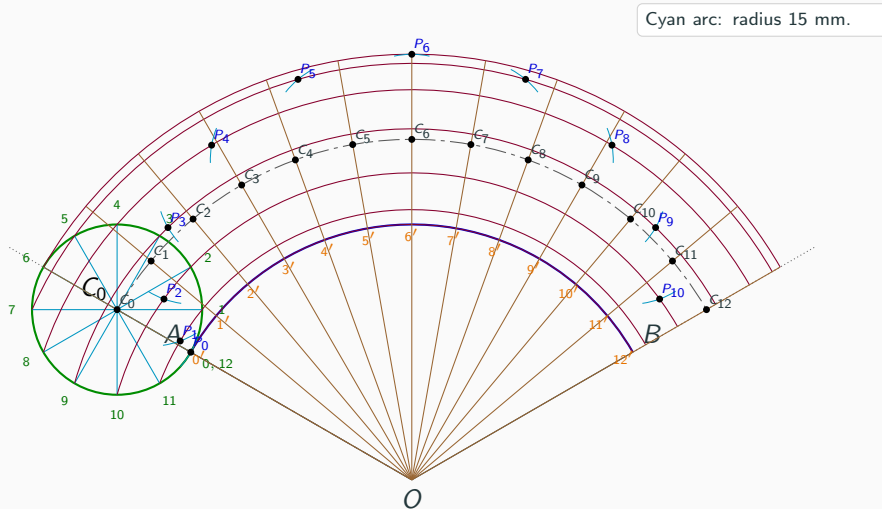


Step 13: Draw the small cutting arc and locate P_i

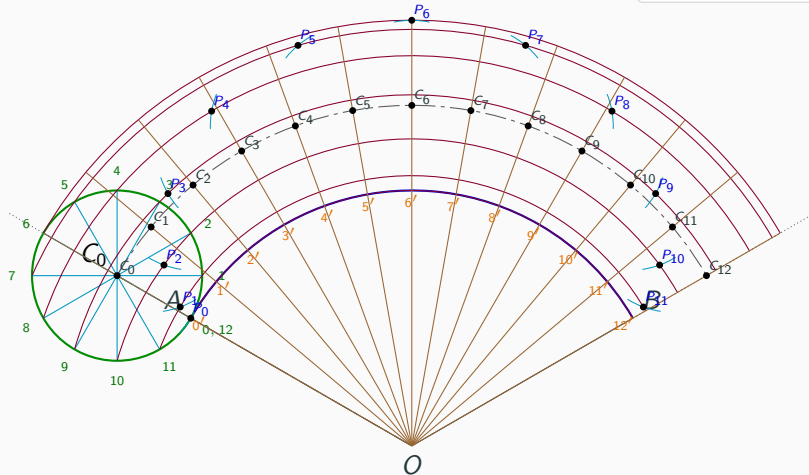


Cyan arc: radius 15 mm.

Step 13: Draw the small cutting arc and locate P_i

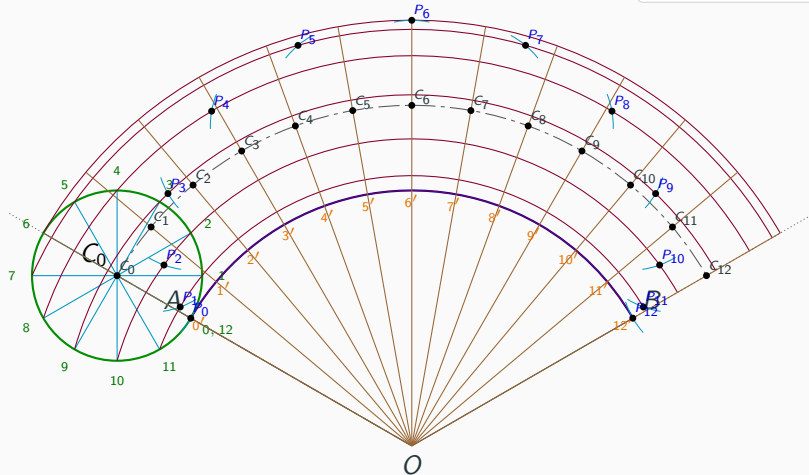


Step 13: Draw the small cutting arc and locate P_i



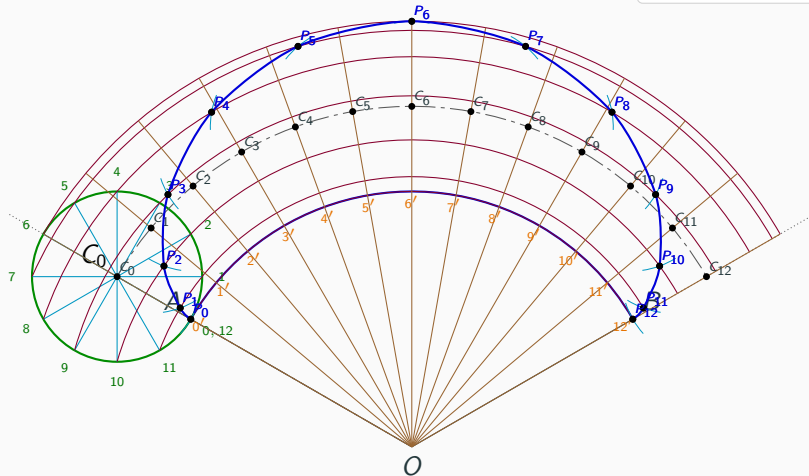
Cyan arc: radius 15 mm.

Step 13: Draw the small cutting arc and locate P_i



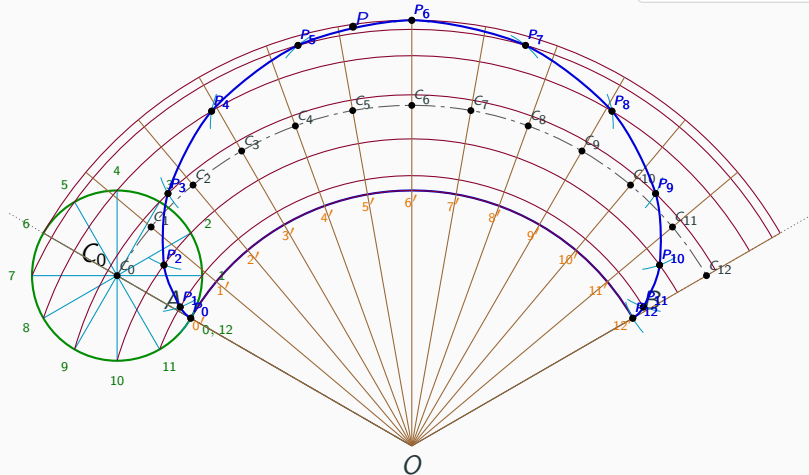
Cyan arc: radius 15 mm.

Step 14: Draw the epicycloid



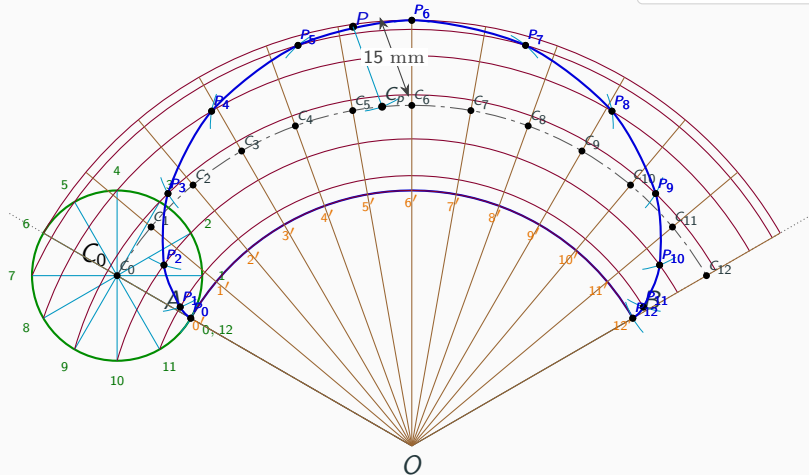
Blue curve: epicycloid.

Step 15: Select a point P on the epicycloid

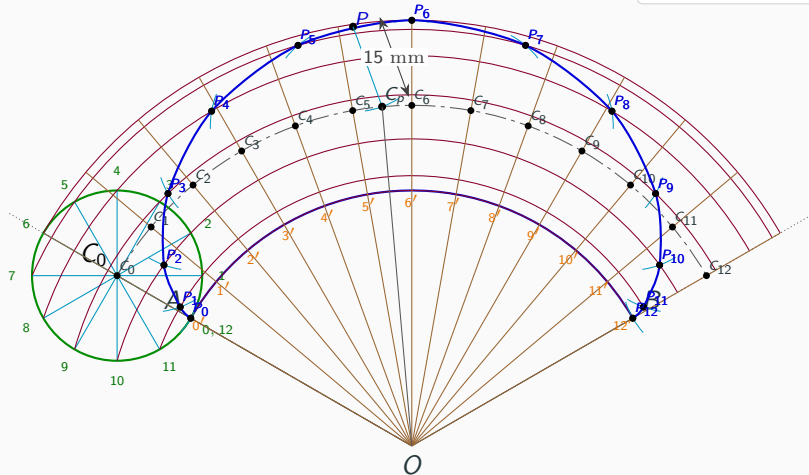


Step 16: Draw the 15 mm arc from P to the centre curve

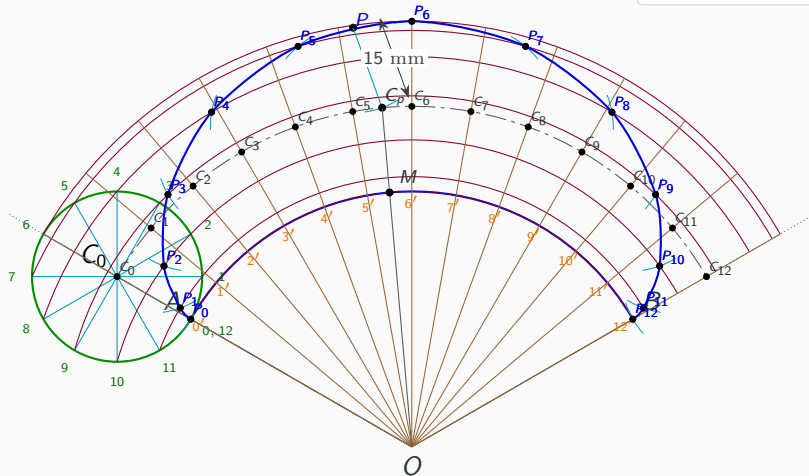
$$PC_P = 15 \text{ mm}$$



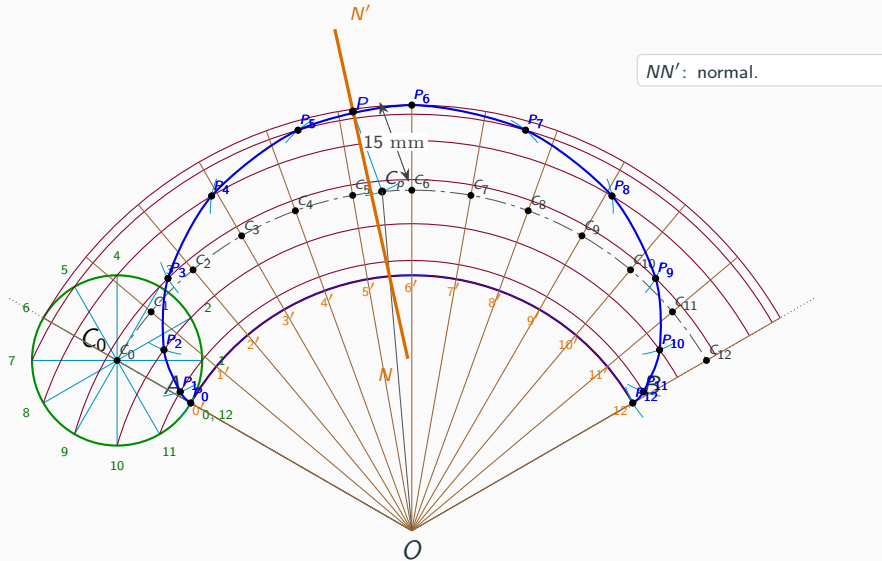
Step 17: Join C_p to O



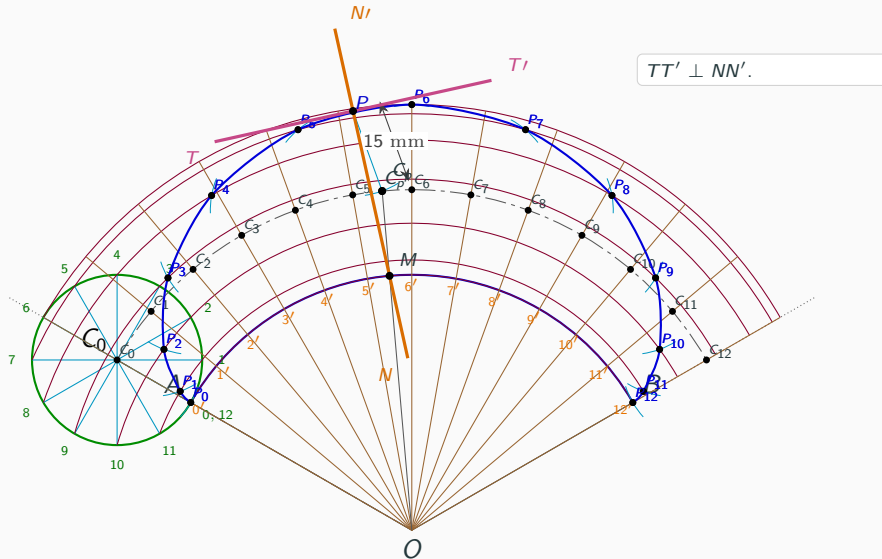
Step 18: Mark M on the directing circle



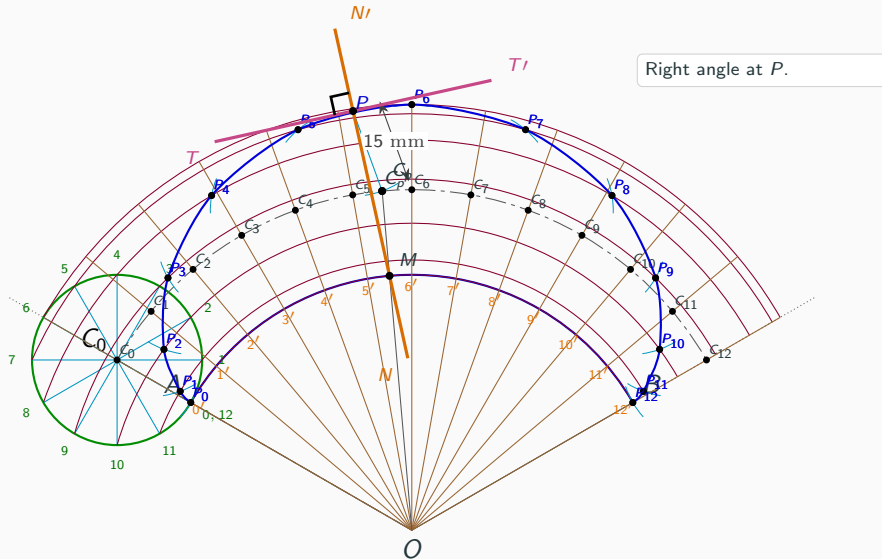
Step 19: Draw the normal NN' through P



Step 20: Draw the tangent perpendicular to NN'



Step 20: Show the right-angle at P



Step 20: Final completed epicycloid construction

